JY063 STIM Rev 1: hand calculations

Worst-case analysis of one stimulator channel, block by block

UBC ELEC 491 Capstone, project JY063
Companion to deck 04 (*Rev 1 design review*) and to `STIM_Rev1_Design/02_calc/`

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Abstract

The Rev 1 channel is a current source. It drives up to ±5 mA into a head of 0.5–13 kΩ from ±100 V rails, inside a chain of protection circuits. Before any circuit simulation, every part was sized by hand from datasheet limits, worst case first. This document writes out those calculations as equations, so that a reader can check each step with pencil and paper. It covers ten blocks, and each block has a Python script and a JSON result file in the repository.

The hand results set the expected value for every simulation in the design review deck. They also produced three findings that changed the design:

1. one fixed limiter resistor cannot cover the DN2540 part spread;
2. the output accuracy fails at 0.5–1 mA without a per-board zero calibration;
3. the limiter cannot survive a continuous double fault in its small package.

One hand model was later shown by simulation to be optimistic: the dynamics of the mismatch trip (§7).

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1 How to read this document

Where the numbers live. Sections 2–11 correspond, in order, to the scripts `01_*.py` to `10_*.py` in `02_calc/`. Each script writes `results/<name>.json`. The command `python3 run_all.py` in `02_calc/` regenerates all of them in about a second. Section 12 compares every result with the simulation and points to the matching slide of deck 04.

Tags. Every input carries a tag:

- [DS] datasheet value (the part and page are in the script);
- [ASSUME] assumption, with a stated reason;
- [DEC] design choice, recorded in `DECISIONS.md`;
- [H] result of another block;
- [N] ngspice result recorded in the repository.

All results of this document are [H].

Worst-case method. Unless stated otherwise, the method is the same throughout:

- Each uncertain input p_i is given an interval $[p_i^-, p_i^+]$.
- The output $f(p)$ is evaluated at every corner of the box $\prod_i [p_i^-, p_i^+]$, i.e. every combination of interval end points, and the minimum and maximum are reported.
- This gives the true extremes when f is monotone in each p_i separately, which holds for every block here. Where it might not, a fine grid is used instead (the DN2540 pinch-off voltage V_P is stepped in 50 mV increments).
- Error budgets (§11) report both the linear worst-case sum $\sum_i |e_i|$ and the root-sum-square $\sqrt{\sum_i e_i^2}$.

The channel in one line. DAC command V_{cmd} → input clamp and buffer → Howland current pump → $R_7 = 1\,\mathrm{k\Omega}$ → current limiter → channel switch → electrode, head, return electrode → return switch → return shunt → ground.

The pump forces $I_{\mathrm{out}} = V_{\mathrm{cmd}}/R_7$, so 1 V of command gives 1 mA. Four monitors watch the channel: over-current on the rail shunts, return mismatch, electrode over-voltage, and rail loss. Any of them sets a latch that opens both switches.

Requirements used.

- *Output accuracy* (KPI 1): $|e| \le \max(1\% \cdot |I|,\ 5\,\mu\mathrm{A})$.
- *Load reach* (KPI 2): 5 mA into $\ge 13\,\mathrm{k\Omega}$ at the worst corner.
- *Single-fault limit* (S1): $\le 10\,\mathrm{mA}$ after $1\,\mu\mathrm{s}$.
- *Limiter window* (S2): $5.5 < I_{\mathrm{lim}} < 10\,\mathrm{mA}$.
- *Trip below clamp* (S3).
- *No nuisance trips* (S4).

Temperature range: 0–50 °C. Rail range: 90–110 V. All are defined in the design brief (`design_brief.md`).

Part change after these calculations. The hand calculations used the CPC1025N PhotoMOS. The parts list later replaced it with the AQY214S, because the CPC1025N is being discontinued. The AQY214S has a lower on-resistance and is faster, so every switch-related number below is conservative. The affected simulations were re-run with the AQY214S.

2 Input clamp and buffer

Script `01_clamp_buffer.py`. Deck 04: "Walking through the channel", "Trip below clamp".

The command passes through $R_{S1} = 1\,\mathrm{k\Omega}$, with two back-to-back 6.2 V zeners to ground, and then through the buffer U3 (OPA171) supplied from $\pm V_b$. A runaway DAC output, $V_{\mathrm{DAC}} = 10\,\mathrm{V}$ [ASSUME], drives one zener into breakdown and the other into forward conduction. The clamp voltage V_{cl} solves the fixed-point equation

$$V_{cl} = V_{Z5} + Z_{ZT}\,(I_Z - 5\,\mathrm{mA}) + \alpha_Z\,\Delta T + V_F + \alpha_F\,\Delta T, \qquad I_Z = \frac{V_{\mathrm{DAC}} - V_{cl}}{R_{S1}}, \tag{1}$$

with:

- $V_{Z5} \in [5.8,\ 6.6]\,\mathrm{V}$ and $Z_{ZT} = 10\,\Omega$ [DS];
- $\alpha_Z \in [0.4,\ 3.7]\,\mathrm{mV/K}$ [DS];
- $V_F \in [0.6,\ 0.8]\,\mathrm{V}$ and $\alpha_F = -2\,\mathrm{mV/K}$ [ASSUME];
- $\Delta T = T - 25\,^{\circ}\mathrm{C}$, with $T \in \{0, 25, 50\}$.

The map is a contraction, because $Z_{ZT}/R_{S1} = 0.01 < 1$, so simple iteration converges. The script evaluates all 24 corners:

$$V_{cl} \in [6.344,\ 7.418]\,\mathrm{V} \implies I_{\mathrm{ceiling}} = V_{cl}/R_7 \in [6.34,\ 7.42]\,\mathrm{mA}. \tag{2}$$

The lower end, 6.34 mA, is the upper bound for the over-current trip level (§6).

Buffer rails ($V_b = 9\,\mathrm{V} \pm 2\%$, decision D-01).

- *Linear range.* The OPA171 is specified to $V_+ - 2\,\mathrm{V}$ [DS]. The upper end of the linear range is therefore $0.98 \cdot 9 - 2 = 6.82\,\mathrm{V}$, which is above the 5.5 V needed for $5\,\mathrm{mA} + 10\%$.
- *Zener open.* If a zener fails open, the buffer output can reach $V_b^+ - 0.03\,\mathrm{V} = 9.18 - 0.03 = 9.15\,\mathrm{V}$, i.e. $9.15\,\mathrm{mA} < 10\,\mathrm{mA}$. This second ceiling is what keeps the single-fault limit S1 when a zener is open, and it is the smallest single-fault margin in the design (0.85 mA).

Zener leakage. The datasheet gives $I_R \le 3\,\mu\mathrm{A}$ at 4 V [DS]. Scaling linearly to 5 V and doubling for 50 °C [ASSUME] gives $I_R = 7.5\,\mu\mathrm{A}$. Through R_{S1} this is a 7.5 mV error on a 5 V command, i.e. -0.15%.

3 Pump load reach (compliance)

Script `02_pump_compliance.py`; model `STIM_compliance_sim/tools/generator/stim_model.py`. Deck 04: "Load reach"; deck 03.

Model. The INA149 and its buffer sit on a *floating* supply, which is bootstrapped to the output by two followers.

- The upper follower's base (node A) is fed from the rail V_r through $R_t = 88.4\,\mathrm{k\Omega}$, and from A through R_3 and diode D_1 to OUT.
- The bottom string mirrors this with current y.

Let x be the current through R_3, β the composite current gain of the follower, I_q the quiescent current of the floating chips, I_L the load current and V_D the diode drop. With every resistor tolerance ϵ taken in the unfavourable direction, Kirchhoff's current law at A gives

$$y = \frac{(V_r + V_{\mathrm{out}} - V_D)/R_t^- - I_q/(\beta+1)}{1 + R_3^-/R_t^-}, \tag{3}$$

$$x = \frac{(V_r - V_{\mathrm{out}} - V_D)/R_t^+ - (I_q + I_L + y)/(\beta+1)}{1 + R_3^-/R_t^+ - 1/(\beta+1)}, \tag{4}$$

with $R^{\pm} = R(1 \pm \epsilon)$. The headroom of the INA149 output to its own supply is

$$H = x\, R_3^{-} + \Delta V, \tag{5}$$

where ΔV is the diode-to-V_{BE} mismatch: $-0.15\,\mathrm{V}$ for a single transistor and $-0.25\,\mathrm{V}$ for a Darlington [ASSUME]. If $x < 0$, the diode turns off and the base is fed through R_t only; the script handles that branch.

The output voltage needed for a load R_L is

$$V_{\mathrm{out}} = I_L(R_L + R_7) + V_{\mathrm{ser}}, \tag{6}$$

where V_{ser} is the drop across the protection chain. The *load reach* is

$$R_{\max} = \sup\{R_L : H(R_L) \;\geq\; H_{\min} = 1.5\,\mathrm{V}\} \qquad \text{(INA149 output swing [DS])}. \tag{7}$$

H decreases monotonically in R_L, so $R_{\max}$ is found by bisection.

Inputs at the worst corner.

- Rail: $V_r = 90\,\mathrm{V}$, reduced by the drop of the rail current $I_q + I_L + I_{\mathrm{string}}$ across the fuse ($0.5\,\Omega$) and the rail shunt ($47\,\Omega$).
- Gain: $\beta = 30$ for the output transistor [DS], multiplied by 26 for a Darlington driver [N].
- Tolerances: $\epsilon = 1\%$.
- Protection-chain drop:

$$V_{\mathrm{ser}} = \underbrace{2.188}_{\text{limiter, §4}} + \underbrace{2 \cdot 37.5\,\Omega \cdot 5\,\mathrm{mA}}_{\text{two PhotoMOS}} + \underbrace{10\,\Omega \cdot 5\,\mathrm{mA}}_{\text{return shunt}} = 2.613\,\mathrm{V}.$$

Output-stage option (deck 03 name)	90 V	100 V	110 V
T1a client board (TLE2141, single transistor, $R_3 = 10\,\mathrm{k}\Omega$)	**6.16**	8.04	9.93
T1c OPA171 only	8.39	10.28	12.17
T2a Darlington only	12.34	14.33	16.33
T2b OPA171 + Darlington + $R_3 = 14.3\,\mathrm{k}\Omega$ (**chosen**)	**13.42**	15.41	17.41

Table 1: $R_{\max}$ in kΩ at the worst corner, with the full protection chain [H].

The hand model is $0.27\,\mathrm{k}\Omega$ optimistic against ngspice at this corner (deck 03, test T2b). Subtracting that offset gives $13.15\,\mathrm{k}\Omega$ for the chosen option, which is the value later confirmed by the full Rev 1 simulation (13.15 at 0 °C, 13.19 at 25 °C).

Floating-supply span. At idle, with $\beta \to \infty$, the span across the floating chips is $2[(V_r - V_D)\,R_3/(R_t + R_3) + \Delta V] = 30.3\,\mathrm{V}$ at 110 V. This is below the OPA171 maximum of 36 V [DS].

4 Current limiter (DN2540)

Script `03_limiter.py`. Deck 04: "Current limiter".

Model. A depletion MOSFET with source resistor R, and its gate tied to the far end of R, limits at the current I that satisfies $V_{GS} = -IR$. In saturation, the square law with pinch-off voltage $V_P = |V_{GS(\mathrm{off})}|$ gives

$$I = K\,(V_P - IR)^2\,(1 + \lambda V_{DS}). \tag{8}$$

Writing $k = K(1 + \lambda V_{DS})$, this becomes the quadratic

$$kR^2I^2 - (2kRV_P + 1)\,I + kV_P^2 = 0, \qquad I_{\mathrm{lim}} = \frac{(2kRV_P + 1) - \sqrt{4kRV_P + 1}}{2kR^2}. \tag{9}$$

The smaller root is the physical one: the larger root has $IR > V_P$, which means the channel is pinched off. The inputs are:

- $K = 0.12\,\mathrm{A/V^2}$ [ASSUME] (a fit to the DN2540 transfer curve), with a part-to-part spread of $\pm 25\%$ [ASSUME];
- $\lambda = 0.002\,\mathrm{V^{-1}}$ [ASSUME];
- $V_P \in [1.5,\ 3.5]\,\mathrm{V}$ [DS];
- $dV_P/dT \in [0,\ 4.5]\,\mathrm{mV/K}$ [DS];
- $V_{DS} \in [5,\ 105]\,\mathrm{V}$;
- R within $\pm 1.25\%$, plus a tempco of $100\,\mathrm{ppm/K}$.

Option A: one fixed R for every part. I_{lim} decreases monotonically in R and increases in V_P. So the lowest current occurs at V_P^- and the highest at V_P^+:

$R\ (\Omega)$	120	150	180	220	270	330
I_{lim} window (mA)	8.9–27.1	7.3–21.9	6.2–18.4	5.2–15.2	4.3–12.5	3.6–10.3

No value satisfies $5.5 < I_{\mathrm{lim}} < 10\,\mathrm{mA}$. The reason: for $kRV_P \gg 1$ the root gives $I_{\mathrm{lim}} \approx V_P/R$, so with a fixed R the limit scales with V_P. The window allows a ratio of $10/5.5 = 1.82$, but the datasheet spread is $3.5/1.5 = 2.33$, before any K, temperature or V_{DS} spread is added. **Fail: a trim is required.**

Option D (chosen, decision D-02): matched pair, one resistor chosen on test.

- Each DN2540 is measured at the target current $I_t = 7.6\,\mathrm{mA}$.
- Two parts whose $|V_{GS}(I_t)|$ agree within $100\,\mathrm{mV}$ [ASSUME] are paired.
- One shared resistor is fitted:

$$R = \mathrm{E96}\left(\frac{|V_{GS}(I_t)| + e_m}{I_t}\right), \qquad |V_{GS}(I_t)| = V_P - \sqrt{I_t/k}, \tag{10}$$

  where $e_m = \pm 10\,\mathrm{mV}$ is the error of the test fixture.

Over the full V_P grid and every corner above, the result is

$$I_{\mathrm{lim}} \in [6.35,\ 8.86]\,\mathrm{mA} \subset (5.5,\ 10)\ \checkmark, \qquad R \in [165,\ 432]\,\Omega. \tag{11}$$

The target $7.6\,\mathrm{mA}$ was chosen to centre this window; the optimum is $7.7\,\mathrm{mA}$, with $1.21\,\mathrm{mA}$ of margin. For comparison, binning parts into N equal V_P bins first passes at $N = 4$ (5.75–9.67 mA).

Voltage drop at $5\,\mathrm{mA}$ (an input to §3). Below the limit, both transistors are in the triode region, where $I = 2K[(V_{GS} + V_P)V_{DS} - V_{DS}^2/2]$. Solving for V_{DS} and adding IR gives a drop of 0.87–2.19 V across the pair. Two separate resistors (option C) would give up to 4.35 V. Sharing one resistor therefore halves the cost in load reach.

Dissipation. With the electrode shorted to a 110 V rail, the limiting transistor dissipates $8.86\,\text{mA} \cdot 110\,\text{V} = 0.975\,\text{W}$ until the switches open (§10).

5 PhotoMOS switches

Script `04_photomos.py`. Deck 04: "Disconnects".

The three switch LEDs, for U4, U5 and U6, are in series from $+12\,\text{V} \pm 5\%$ through R_{LED} and a 2N7002 transistor, whose on-resistance R_{DS} is at most $7.5\,\Omega$. The LED current is

$$I_F = \frac{V_{\text{LED}} - 3V_F}{R_{\text{LED}} + R_{DS}}. \tag{12}$$

With $R_{\text{LED}} = 931\,\Omega$ and $V_F \in [0.9, 1.5]\,\text{V}$ [DS], the two extreme corners are:

- weak corner: $I_F = (11.4 - 4.5)/938.5 = 7.35\,\text{mA}$, which is at least the 5 mA test condition of the datasheet;
- strong corner: $I_F = (12.6 - 2.7)/931 = 10.6\,\text{mA}$, which is at most 20 mA.

Charge after a trip. Suppose a fault is one that only the switch can stop, such as an output stage shorted to a rail. The limiter then holds $I_{\text{lim,max}}$ until the switch opens, so the charge delivered is

$$Q = I_{\text{lim,max}}\,(t_{\text{detect}} + T_{\text{off}}) = 8.86\,\text{mA} \cdot (30\,\mu\text{s} + 1\,\text{ms}) = 9.13\,\mu\text{C}$$

for the CPC1025N. With the AQY214S (decision D-17), the simulation gives $1.8\,\mu\text{C}$.

6 Over-current trip

Script `05_trip_oc.py`. Deck 04: "Single faults", "Nuisance trips".

Sensing. Each rail carries a shunt R_{sh}. Two INA149 amplifiers are chained so that

$$V_{21} = V_{\text{MID}} + (I_P - I_N)\,R_{sh} = V_{\text{MID}} + I_{\text{load}} R_{sh},$$

in which the quiescent currents, common to both rails, cancel. The signal is then amplified by $G = 4$, filtered by two equal RC poles ($\tau = 22\,\mu\text{s}$), and compared against a window at $V_{\text{MID}} \pm G\,I_{\text{trip}}\,R_{sh}$.

The rail shunts are used, rather than R_7, because a short from the output stage to a rail bypasses R_7.

Allowed window for I_{trip}. The two-pole filter has impulse response $h(t) = (t/\tau^2)\,e^{-t/\tau}$. Its maximum is at $t = \tau$:

$$\max_t h(t) = h(\tau) = \frac{1}{e\,\tau}.$$

An overshoot of charge Q_{os} that is short compared with τ therefore adds at most $Q_{os}/(e\tau)$ to the filtered current. With $Q_{os} = 4\,\text{nC}$ [ASSUME] (a 2 mA, 2 μs overshoot) and a worst-case accuracy of 0.75% (§11), the filtered normal peak is

$$I_{\text{norm}} = 5\,\text{mA}\,(1.0075) + \frac{4\,\text{nC}}{e \cdot 22\,\mu\text{s}} = 5.104\,\text{mA}. \tag{13}$$

The trip level must sit above this peak, so that normal use never trips (S4), and below the lowest clamp ceiling of §2, so that a runaway command always trips (S3). With 0.05 mA of margin on each side, the allowed window is $[5.154,\ 6.294]\,\text{mA}$. The trip level is set at its centre, $I_{\text{trip}} = 5.724\,\text{mA}$.

Sensing error as a function of R_{sh}. Every error source is converted to an equivalent current:

- two INA149 offsets, their drift over 25 K, and their common-mode rejection at 110 V, all divided by R_{sh};
- the input-current imbalance when the two rails differ by 20 V: 20 V/400 kΩ;
- the offsets of the gain stage and of the comparator;
- the error of the threshold-resistor ladder;
- the tolerance of R_{sh};
- rail ripple (0.29 V_{pp} at 150 kHz) after the two filter poles.

R_{sh} (Ω)	10	22	33	**47**	68	100
total error (mA)	1.37	0.71	0.53	**0.42**	0.34	0.29
fits the window?	**no**	**no**	yes	**yes**	yes	yes
rail drop (V)	0.08	0.17	0.26	**0.36**	0.53	0.78

A value fits when $I_{\text{trip}}\pm$ (total error) stays inside the window. The INA149 common-mode rejection term dominates and scales as $1/R_{sh}$. The chosen 47 Ω (decision D-06) gives an effective trip range of 5.30–6.14 mA, which is inside [5.154, 6.294].

Trip time. For a current step I_f, the filtered signal is

$$y(t) = I_f\big(1 - (1 + t/\tau)\, e^{-t/\tau}\big).$$

The trip time is the root of $y(t) = I_{\text{trip}} + e_{\text{tot}}$, found by bisection:

- 45.7 µs for a 10 mA fault;
- 117 µs for a runaway at the lowest clamp ceiling, where the overdrive above the trip level is only 0.2 mA.

During that time the current is bounded by the clamp at ≤ 7.42 mA, so S1 holds. The simulation gives 65–78 µs.

7 Return-mismatch trip

Script `06_trip_mismatch.py`. Deck 04: "Nuisance trips", "Three methods agree".

The trip compares the current that comes back through the return shunt, i_r, with the commanded current, i_c, and fires if the filtered difference leaves a band $\pm b$. During a ±5 mA square wave the two differ for a few microseconds, for three reasons:

- the command slews at the buffer's slew rate, 1.5 V/µs;
- the pump responds as a second-order system,

  $$\ddot{i}_o = \omega_0^2\,(i_c - i_o) - 2\zeta\omega_0\,\dot{i}_o, \qquad f_0 = 0.85\,\text{MHz},\ \zeta = 0.22,$$

  with f_0 and ζ taken from the measured ringing of the client board;
- part of the output current charges the capacitance C_g from the electrode to ground:

  $$i_r = i_o - C_g\,\frac{d(i_o R_L)}{dt}, \qquad C_g \in \{70, 150\}\,\text{pF}\ \text{[ASSUME]}.$$

The error $e = i_r - i_c$ is integrated numerically with a step of 2 ns and passed through the two-pole filter. The band is then set to

$$b = \lceil 2 \cdot \text{peak} + e_{\text{static}} \rceil_{0.05\,\text{mA}},$$

where $e_{\text{static}} = 80.4\,\mu\text{A}$ is the sum of the amplifier, resistor, leakage and comparator errors.

τ	4.7 μs	10 μs	22 μs	**47 μs**
filtered peak (mA)	1.47	0.73	0.34	**0.16**
band b (mA)	3.05	1.55	0.80	**0.40**
time to detect a lost return (μs)	9.7	11.2	15.7	**21.9**

A longer τ trades detection time for a narrower band. The design chooses $\tau = 47\,\mu$s and $b = 0.40\,$mA (decision D-07); a lost return is detected in 22 μs, within the 50 μs target.

Where this hand model was wrong. In ngspice, with the real pump, the filtered peak at 13 kΩ is 0.36–0.39 mA, not 0.16 mA. The second-order pump model underestimates the lag between the commanded and returned currents. The fix was to delay the command reference by 3.92 μs (revision of decision D-07), which brings the peak back to 0.16 mA and restores a margin of 0.24 mA. This is the only hand model in this document that simulation overturned. The mismatch band must therefore be tuned from simulation, not from this section.

8 Electrode over-voltage trip

Script `07_trip_ov.py`.

A divider $R_{\text{top}} = 20\,\text{M}\Omega$, $R_{\text{bot}} = 200\,\text{k}\Omega$, referenced to V_{MID}, scales the electrode voltage by

$$k = R_{\text{bot}}/(R_{\text{top}} + R_{\text{bot}}) = 1/101,$$

so an 80 V threshold becomes $\delta = 80/101 = 0.792\,$V. The errors, referred to the electrode, are:

- divider ratio ($2 \times 1\%$): 1.60 V;
- comparator offset, 5.5 mV/k: 0.56 V;
- comparator bias current through the divider's output resistance: 0.40 V;
- threshold ladder: 0.41 V.

The trip level therefore lies in [77.0, 83.0] V. This is above the 65 V of normal use (5 mA $\times$ 13 kΩ) plus 5 V of margin, and below the 111 V minimum breakdown of the surge-protection diode (TVS). The filter time constant is $(R_{\text{top}} \parallel R_{\text{bot}})\,C = 198\,\text{k}\Omega \cdot 47\,\text{pF} = 9.3\,\mu$s.

9 Fault latch, enable logic and timing

Script `08_latch_enable.py`. Deck 04: "What changed" (latch reset).

The logic is small enough to verify exhaustively. The script enumerates all $2^6 = 64$ combinations of six inputs (trip, power-on reset, FPGA reset, enable request, rail-good, previous latch state). For each, it asserts the safety property

$$\text{LED drive on} \implies \neg\text{trip} \wedge \text{rail-good} \wedge \text{enable} \wedge \neg\text{latched}.$$

- Chosen design (decision D-10): reset through the clock input with $D = 0$, so that a preset always dominates. **0 violations.**
- Rejected variant: reset through the clear input. **6 violations**, because the '74 flip-flop with preset and clear both active drives $Q = \bar{Q} = 1$, which would enable the LEDs.

The logic delay is 0.62 μs. The time from a fault to an open switch is therefore set by the switch: about 1.05 ms with the CPC1025N used here, and 0.24 ms with the AQY214S, from simulation. The faster path, clearing the DAC, brings the current to zero in 34 μs.

10 Power and junction temperature

Script `09_power_thermal.py`.

Each junction temperature is $T_j = T_a + P\,\theta_{JA}$, with $T_a = 50\,°\mathrm{C}$ and $V_r = 110\,\mathrm{V}$:

- Output transistor Q2: $P = 0.621\,\mathrm{W}$ (from the bootstrap model, worst at $V_{\mathrm{out}} = 0$ while sourcing), $\theta_{JA} = 80\,\mathrm{K/W}$ [DS]. $T_j = 99.7\,°\mathrm{C} < 125$. **Pass.**
- Limiter in a single fault: 0.975 W for at most 1 ms, about 1 mJ. **Pass** (transient).
- Limiter in a continuous double fault (both switches welded):
  - small SOT-89 package: $50 + 0.975 \cdot 133 = 180\,°\mathrm{C} > 150$ **(fail)**;
  - TO-220 package: 78 °C.

  This is a finding for the layout and for the double-fault discussion. It is outside the single-fault rule.

11 DC error budget (output accuracy)

Script `10_error_budget.py`. Deck 04: "Output accuracy".

For an output current I into a load R_L, the electrode voltage is $V_e = IR_L$ and the sense node sits at $V_s = V_e + 2.6\,\mathrm{V}$. With a temperature excursion of $\Delta T = 25\,\mathrm{K}$ from 25 °C, the error terms are:

$$e_{R7} = I\,(\epsilon_{R7} + \alpha_{R7}\Delta T), \qquad e_{\mathrm{gain}} = I\,(\epsilon_G + \alpha_G \Delta T),$$
$$e_{\mathrm{off},j} = (V_{os,j} \cdot [\text{not calibrated}] + \dot{V}_{os,j}\,\Delta T)/R_7, \qquad e_{\mathrm{CMRR}} = 10^{-90/20}\,V_s/R_7,$$
$$e_{\mathrm{zener}} = 0.15\% \cdot I, \qquad e_{\mathrm{leak}} = V_e/20.2\,\mathrm{M\Omega} + V_e/10\,\mathrm{M\Omega} + V_s/400\,\mathrm{M\Omega}.$$

Here j runs over the three amplifiers: INA149 (1.1 mV), U1 and U3 (OPA171, 1.8 mV each). The limit is $\max(0.01\,I,\ 5\,\mu\mathrm{A})$.

	as built		with zero calibration	
point	worst sum	RSS	worst sum	RSS
0.5 mA, 0.5 kΩ (limit 5 µA)	**7.10**	3.21	2.40	1.19
0.5 mA, 13 kΩ (limit 5 µA)	**8.24**	3.30	3.54	1.42
1 mA, 13 kΩ (limit 10 µA)	**11.23**	4.06	6.53	2.75
5 mA, 13 kΩ (limit 50 µA)	35.07	13.91	30.37	13.59

All values are in µA.

- *As built*, the three offsets add $(1.1 + 1.8 + 1.8)\,\mathrm{mV}/1\,\mathrm{k\Omega} = 4.7\,\mu\mathrm{A}$ before any drift, which nearly uses the whole 5 µA allowance at 0.5 mA. The worst-case sum fails at 0.5–1 mA, with a worst margin of $-3.24\,\mu\mathrm{A}$. The RSS passes.
- *With a one-point zero calibration* stored per board by the FPGA (decision D-12, release condition 1), the offsets are removed and only their drift remains. Every point passes, with a worst margin of $+1.46\,\mu\mathrm{A}$.

The simulation confirms the pattern: uncalibrated, 12 of 72 points fail at ±0.5 mA; calibrated, the worst error is 0.31 of the limit.

12 Hand calculation against simulation

The full comparison is `04_compare/tables.md`: 21 tables and 131 rows, with every difference above 5% explained.

Quantity	Hand [H]	ngspice [N]	Comment, deck 04 slide
Load reach, worst corner	13.42 (13.15 with offset) kΩ	13.15 kΩ	Agree within 2 %. "Load reach"
Limiter window, trimmed pair	6.35–8.86 mA	6.43–8.80 mA	Hand range is slightly wider (coarser corners). "Current limiter"
One fixed limiter resistor	fails	fails (4.43–10.81 mA)	Same conclusion; also an LTspice test
Clamp ceiling	6.34–7.42 mA	within 1 %	"Three methods agree"
Over-current trip time	46–117 μs	65–78 μs	Hand value is a deliberate worst case. "Single faults"
Clamp zener open	9.15 mA	not simulable	The OPA171 stand-in model clips at 7.2 V; the hand value is the evidence. "Results that changed"
Mismatch peak, 13 kΩ, no delay	0.16 mA	0.36–0.39 mA	**Hand model 2.2× optimistic**; fixed by a 3.92 μs delay. "Nuisance trips"
Accuracy, uncalibrated	fails at 0.5–1 mA	12/72 fail at 0.5 mA	Same conclusion. "Output accuracy"

What a reviewer should check first.

1. The bootstrap model of §3: it sets the load reach, the result with the thinnest margin (1.1%).
2. The square-law limiter of §4: K and its spread are assumptions, not datasheet values.
3. The $1/(e\tau)$ argument and the trip window of §6: it fixes R_{sh} and I_{trip}.
4. The exhaustive logic check of §9.

Reproduce

```
cd STIM_Rev1_Design/02_calc
python3 run_all.py        # all blocks; -v prints everything
cat results/summary.json  # every check, PASS or FAIL
```

The scripts need Python 3 and numpy. Datasheet sources are listed in `STIM_Rev1_Design/01_datasheets/`.